

Assignment 4

» Section A

1. What is TCP?

Ans: TCP stands for Transmission Control Protocol.

It is the main protocol of the Internet protocol suite. It lies between the Application and network layer which are used in providing reliable delivery services. It is connection oriented protocol for communication that helps in exchange of message between different devices over a network.

2. Define UDP?

Ans: UDP stands for User Datagram Protocol is a transport layer protocol. UDP is a part of Internet protocol suite, referred to as UDP/IP suite. Unlike TCP it is an unreliable and connectionless protocol. There is no need to establish a connection prior to data transfer. The UDP helps establish low-latency and loss-tolerating connection establish over a network.

3. Define the term congestion control?

Ans: Congestion Control is a mechanism that control the entry of data packets into the network, enabling a better use of a shared network infrastructure and avoiding congestive collapse. Congestive Control Avoidance Algorithm (CAA) are implemented at the TCP layer as the mechanism to avoid congestive collapse in a network.

4. What is Intra domain Routing.

Ans: Intra domain routing algorithm works only within domain. It need to know only about other routers within their domain. Protocols used in intradomain routing are known as interior-gateway protocol. In this routing, routing takes place within an autonomous network. Intra domain routing protocol ignores the Internet outside the autonomous system (AS). Some of the popular protocol of this routing are RIP and OSPF.

5. Define MAC.

Ans: To communicate or transfer data from one computer to another, we need an address. In computer network various types of addresses are introduced; each works at a different layer. A MAC address, an unique of 48-bit hardware numbers of a computer that are embedded into a network card known as (Network Interface Card). It is a physical address that works at the Data link layer.

>> Section B

1. What is Transport layer? Explain Reliable and Unreliable in transport layer?

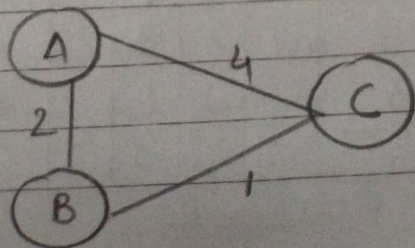
Ans: The Transport layer is layer 4 of the open System Interconnection (OSI) model. It is responsible for ensuring that the data packet arrive accurately and reliably between sender and Receiver. The Transport layer most often use TCP or UDP. In TCP/IP model, transport layer comes between the application

and network layer.

- Reliable Connection :- For reliable connection, it uses TCP. The TCP Protocol uses a three-way handshake, windowing and sequence numbers for guaranteed data delivery.
- Unreliable Connection :- The main problem with reliable connection is that to establish it, TCP always goes through a handshake process before data can be transferred and then acknowledged, that all data sent was received. If you want to send only one piece of information and get a single reply back, going through a handshake process is worthless. It adds overheads & delays.

2. Explain Distance vector routing with diagram.

Ans: A distance vector Routing (DVR) protocol requires a router inform its neighbours of topology changes periodically. Historically it is known as old ARPNET routing algorithm or Bellman ford algorithm. In DVR, each router maintains a routing table. It contains only one entry for each router. It contains two parts - a preferred outgoing lines to use for that destination and an estimate of time (delay). Tables are updated by information with the neighbour's nodes.



In this diagram there are 3 routers A, B, C with following weight $AB=2$, $BC=1$ and $CA=4$

- Step 1: In this, each router shares its routing table with every neighbour.

form A	A	B	C
A	0	2	3
B			
C			

form B	A	B	C
A			
B	2	0	1
C			

form C	A	B	C
A			
B			
C	3	1	0

- Step 2: If the path via a neighbour has a lower cost, then the router updates its local tables to forward packet to the neighbour.

form A	A	B	C
A	0	2	3
B			
C			

form B	A	B	C
A			
B	2	0	1
C			

form C	A	B	C
A			
B			
C	3	1	0

- Step 3: The final updated routing table where lower cost distance vector routing protocol for all routers

form A	A	B	C
A	0	2	3
B	2	0	1
C	3	1	0

form B	A	B	C
A	0	2	3
B	2	0	1
C	3	1	0

form C	A	B	C
A	0	2	3
B	2	0	1
C	3	1	0

Router A

Router B

Router C

3. Explain Efficiency consideration in detail.

Ans:- Efficiency in computer networks involves optimizing the efficiency of utilization of resources to ensure smooth and timely data communication.

Factors like bandwidth utilization, response time and scalability play crucial roles.

Efficient protocols, routing algorithms and error-handling mechanisms contribute to a network's overall effectiveness. It is a balance between performance and resource consumption aiming to deliver optimal result while minimizing delays and maximising throughput.

>> Section C

1. Differentiate between TCP and UDP Protocol.

TCP	UDP
1) Full form: Transmission Control Protocol.	Full form: User Datagram Protocol.
2) Requires an established connection before transmitting the data.	No connection is needed to start and end a data transfer.
3) Can sequence data (send in specific order)	Cannot sequence or arrange data.

- | | |
|--|---|
| 4) can Retransmit data if packet fail to arrive. | No data retransmitting. data can't be received. |
| 5) Delivery is guaranteed. | Delivery is not guaranteed. |
| 6) Broadcasting is not supported. | Broadcasting is supported in UDP. |
| 7) Acknowledgment segment is present. | Acknowledgment segment is not present. |
| 8) Speed is comparatively slower than UDP. | UDP is faster, simpler and more efficient than TCP. |
| 9) TCP is heavy weight | UDP is light weight |
| 10) TCP connection is a byte stream. | UDP connection is a message stream. |
| 11) Protocols used by TCP are HTTP, FTP, SMTP etc. | Protocol used by UDP are RIP, TFTP, DNS etc. |

2. Describe wireless network 802.11 in MAC.

Ans: The 802.11 MAC sublayer provides an abstraction of the physical layer to the logical link layer control or upper layer of the OSI network. It is responsible for encapsulating frames and describing frame formats.

- frame control - It is a 2 bytes starting field composed

of 11 subfields. It contains control information of the frame.

- **Duration:** It is a 2 byte field that specifies the time period for which the frame and its acknowledgment occupy the channel.
- **Address fields:** There are three 6 byte addressess field containing addresses of source, immediate destination and final end point respectively.
- **Sequence:** It is 2 byte field that stores the frame numbers.
- **Data:** This is a variable sized field carries the data from the upper layer. The maximum size of data is 2312 bytes.
- **Check Sequence:** It is a 4 byte field containing error detection information.

Frame control	Duration	Address 1	Address 2	Address 3	Sequence	Data	Sequence check
2byte	2byte	6byte	6byte	6byte	2byte	0-2312 bytes	4byte

IEEE 802.11 MAC Sublayer Frame

3. Explain Congestion Control Algorithm.

Ans: Congestion Control is a mechanism that controls the entry of data packets into the network, enabling a better use of a shared network infrastructure and avoiding congestive collapse. Congestive Avoidance Algorithm (CAA) are implemented at the TCP layer as the mechanism to avoid congestive collapse in a network. There are two types of congestive control Algorithm:-

1. Leaky Bucket Algorithm: The leaky bucket algorithm discover its use in the context of network traffic shaping or rate limiting.

This algorithm is used to control the rate at which traffic is sent to the network and shape the burst traffic to a steady traffic stream. The disadvantage compared with the leaky bucket algorithm are inefficient use of available network resources.

2. Token Bucket Algorithm:- It is a control algorithm that indicates when traffic should be sent. This order comes based on the display of tokens in the bucket. The bucket contains tokens. Each of the token defines the packet of predetermined size. Tokens in the bucket are deleted for the ability to share a packet. When tokens are shown, a flow to transmit traffic appears in display of tokens.